I/O List & Schematic representation of Battlebot

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This document shows the inputs and outputs of the components used in the line following battlebot.

Input

Front ultrasonic sensor

Echo pin → Digital IO 7

Analogue line follower sensors

Line sensor 1 → Analog A0
Line sensor 2 → Analog A1
Line sensor 3 → Analog A2
Line sensor 4 → Analog A3
Line sensor 5 → Analog A4
Line sensor 6 → Analog A5
Line sensor 7 → Analog A6
Line sensor 8 → Analog A7

Output

Left Motor

Motor pin A1 → Digital IO (~PWM) 10 (Left motor back direction)
Motor pin A2 → Digital IO (~PWM) 9 (Left motor straight direction)
Motor pin B1 → Digital IO (~PWM) 6 (Right motor back direction)
Motor pin B2 → Digital IO (~PWM) 5 (Right motor straight direction)

Front ultrasonic sensor

Trigger pin → Digital IO 8

Gripper

GR pin → Digital IO 12

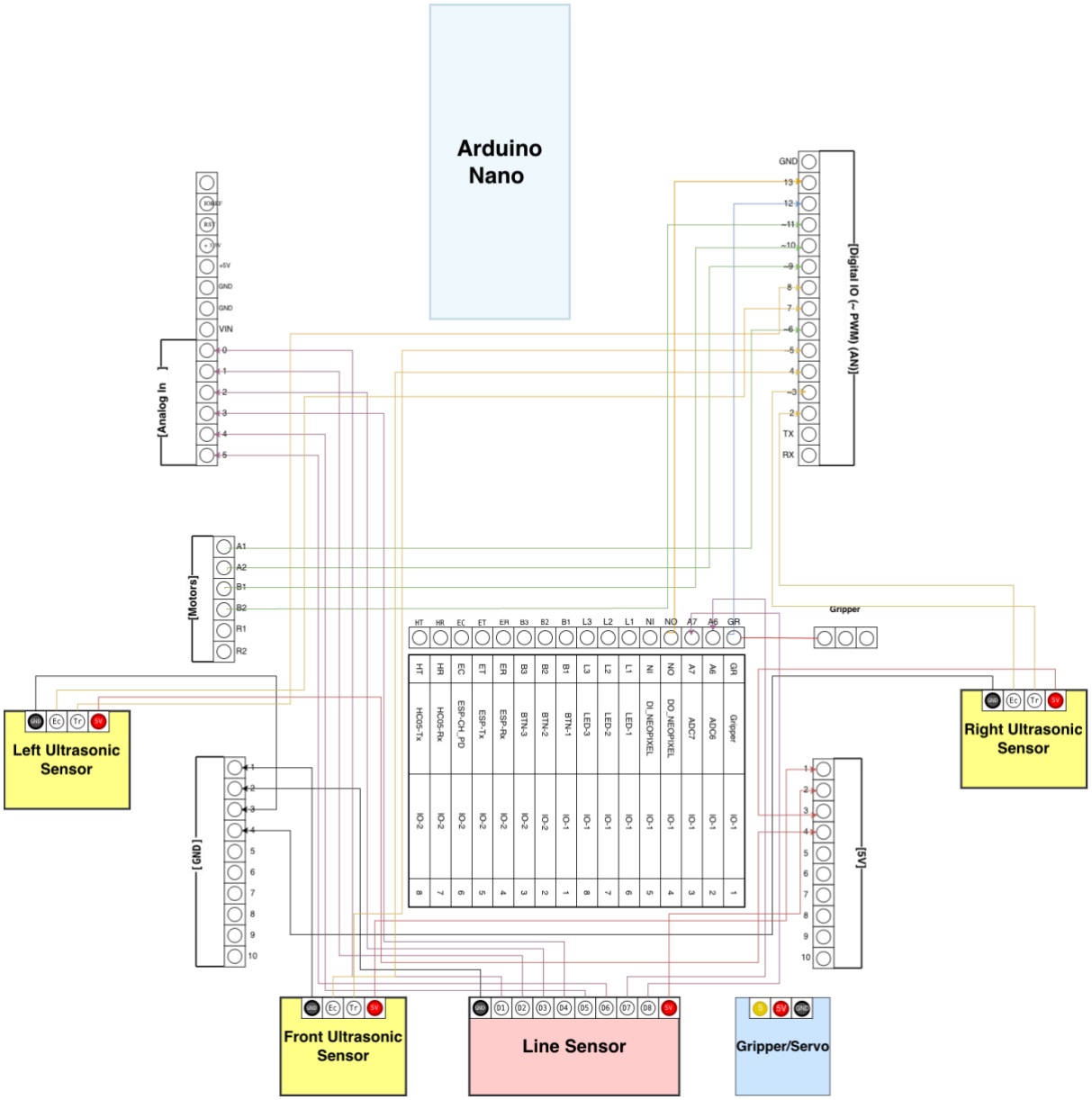
Neopixel

NI pin → Digital IO 13

GND

1. Front ultrasonic sensor
2. Analogue line follower sensor

5V

1. Front ultrasonic sensor
2. Analogue line follower sensor

Schematic representation of Battlebot